

WASP-49b

R-1170

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-49_161113 · created 2026-07-31T03:16:28+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457703.0615 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.062 +/- 0.041
Transit depth (Rp/Rs)^2	0.38 +/- 0.5 [%]
Orbital Inclination (inc)	88.98 +/- 2.87
Airmass coefficient 1 (a1)	25.18 +/- 11.23
Airmass coefficient 2 (a2)	-1.14 +/- 0.83
Scatter in the residuals of the lightcurve fit is	28.7 %
Best Comparison Star	#1 - [629, 263]
Optimal Aperture	2.17
Optimal Annulus	5.51
Transit Duration (day)	3e-06 +/- 8.1e-05

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.062 $\pm$ 0.041 vs 0.116 (deviation 1.32 σ)
Duration fit vs published	nan d vs 0.09 d (deviation nan σ)
Mid-time O-C	-4003.9 min
Depth SNR	1.5
Residual scatter	28.7 %

Light curve

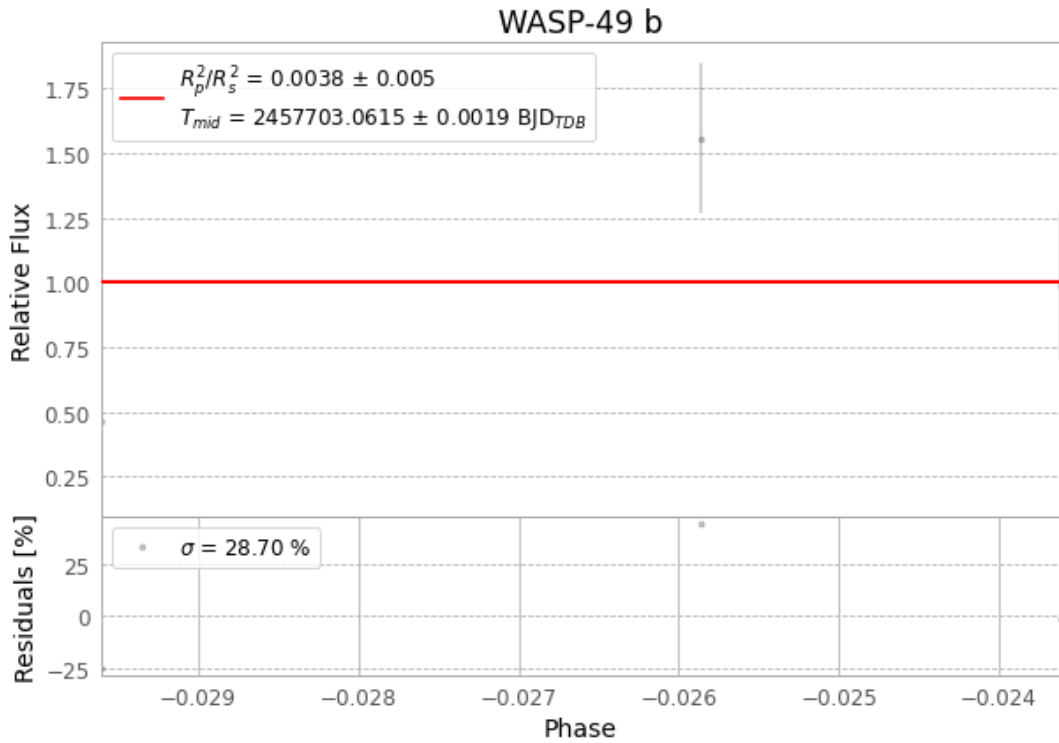


Figure 1 — FinalLightCurve_WASP-49 b_2016-11-12.png (SHA-256 73fc37fadb11673...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [176, 216], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 77bc43d8b0838943...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

104 FITS frames (68 MB), WASP-49161113060912.FITS → WASP-49161113111812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-49-161113.log (7b7c1f10a6e5...), FinalLightCurve_WASP-49 b_2016-11-12.png (73fc37fadb1...), FinalLightCurve_WASP-49 b_2016-11-12.csv (fc4846c35e93...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 03:16Z.